
PrimeCore Minerals

Company Network Hardware Standards

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Table of Contents

Introduction	2
Purpose	2
Router and Firewall.....	3
LAN Switches	4
Office Layout	5
Switches per office	5
Access Points (APs)	6
End Devices	7
Additional Hardware for Network devices.....	7
Hardware and Licence Costs.....	7
Summary Tables	9
Reference.....	11
Appendix A	12
Appendix B	16
Cisco Secure Firewall 1220CX Software Features	17

Introduction

Prime Core Minerals is a mining company with about 600 manual and non-manual employees. With about 200 full-time non-manuals, plus manual workers, visitors, sub-contractors and other people visiting the offices, the company has requested assistance with their network infrastructure setup as each office currently has various Internet Service Providers (ISPs), routers, switches and Access Points (APs) brands and configurations between its offices in Brisbane and each office at site. This has contributed to an unreliable and insecure network and contributed to frustrated staff.

This document outlines the hardware standards that will be implemented at all Prime Core Minerals offices so staff can have a reliable and consistent network at all their offices and visitors can also utilize the company's visitor network without comprising any of the companies' data.

As the company is a mining company, reliability and consistency is a major factor when choosing which brand to use as there isn't always an I.T. expert at site. As this is a major factor in hardware choices and with Cisco being among the top leaders in relation to network hardware over time and trusted by many network engineers, they have been chosen as the hardware brand to recommend for Prime Core Minerals.

To showcase how we would setup their network, we have stood up a virtual environment in EVE-NG 6.2.0-4 and setup a virtual environment of their company. Due to Cisco only licencing Cisco virtual devices on their virtual platform – Cisco Modelling Labs (CML) – we legally and ethically can't use Cisco virtual devices to demonstrate our network design. The Cisco virtual end user licence agreement can be read in full at <https://learningnetwork.cisco.com/s/article/virl-end-user-license-agreement>.

To demonstrate our network design, we have used open-source software that is also free to use. Our base virtual image is Linux Server 26.04, which will have Open vSwitch 3.71 (OVS) installed on top of it to symbolise our network equipment.

Purpose

OVS is a simple virtual image that relies on additional packages to be installed alongside it and therefore the end user must know what features they need to add to secure the switch in a similar way to a Cisco switch. This enables us to show and demonstrate our understanding of network design and security.

The network hardware outlined in this artefact include the SD-WAN, firewall, switches and Access Points (APs).

Router and Firewall

The recommended SD-WAN router for each location is the Cisco Catalyst 8300-1N1S-6T router is in the following link followed by a picture of the device in figure 1.

<https://www.cisco.com/c/en/us/products/collateral/routers/catalyst-8300-series-edge-platforms/datasheet-c78-744088.html#Platformdetails>



Figure 1 – Cisco Catalyst 8300-1N1S-6T

This router hardware specifications would be intensive on our virtual environment as it has 8GB RAM with 16GB M.2 SSD storage with an IPsec throughput of 1.9Gbps so we have utilized the IPsec IKEv2 feature of it through pfSense to connect our 2 sites securely. As pfSense can also act as a firewall, we have taken advantage of that to save the resource consumption in our virtual environment. The recommended firewall is the Cisco Secure Firewall 1220CX as physically it has 8 x 1Gbps RJ45 ports plus 2 x 1/10Gbps ports with a 9Gbps IPsec and threat defence throughput capability. As our virtual environment aren't near these speeds, no limits were placed on the pfSense router/firewall. PfSense does match the 1220CX firewall as they both can handle stateful packet inspection, NAT, VLAN segmentation and VPN support and are therefore a great way to demonstrate how secure their network will be with this hardware.

The datasheet for the Cisco Secure Firewall 1220CX is in the following link followed by a picture of the device in figure 2.

<https://www.cisco.com/c/en/us/products/collateral/security/firewalls/secure-firewall-1200-series-ds.html>



Figure 2 – Cisco Secure Firewall 1220CX

LAN Switches

The recommend switches to use are the Cisco Catalyst C9300-24UX as Cisco switches have proven to be reliable and secure in any office or mine site location in Australia.

The data sheet for the Cisco Catalyst C9300-24UX is in the following link followed by a picture of the device in figure 3. <https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9300-series-switches/nb-06-cat9300-ser-data-sheet-cte-en.html#Platformdetails>



Figure 3 – Cisco Catalyst C9300-24UX

The recommended setup includes the C9300-NM-8X module with a dual power supply, part number: PWR-C1-1900WAC-P

This switch and setup have the following features:

- 24 ports at 10G/5G/2.5G/1G/100M multigigabit port with UPOE available on each port
- 8 x 10G/1G SPF+ ports
 - o allows for fibre connections between buildings. The exact connection method used between switches in different offices is out of scope and it is assumed that it is reliable and configured correctly.
- Dual power supply in case 1 fails or stops receiving power
 - o normal switch setup includes 1 power supply connected to UPS via a power rail and the other power supply connected to general power via a different power rail however for this project and artefact, the power setup of the switches is out of scope and it will be assumed they will be setup this way
- Each power supply can supply 1900W which enables UPOE for each port
- Stackable optional via Stackwise-480
 - o 480Gbps stackable cable with lengths of 0.5m, 1m & 3m available
- 640Gbps switching capacity if solo switch
 - o 1120Gbps switching capacity if stacking
- 476.19Mpps forwarding rate if solo switch
 - o 833.33Mpps forwarding rate if in stackable mode

Office Layout

Switches per office

The number of switches and the exact port allocation can vary per office as there are multiple factors that can influence this. Some of these factors include

- Number of physical servers in the office
- Number of LAN desks required in the office (engineering or I.T. office will benefit more from LAN connections compared to a manager's office which will mainly be laptop users)
- Number of meeting rooms and video conferencing rooms per office
- Size of office and its layout
- Number of desks per office
- Number of external devices (e.g. IoT devices, CCTV cameras) connected to that office

These variations can change so to assist with a standard layout the following guidelines and assumptions are made:

- Each employee will have the ability to have up to 3 connections per allocated desk as this allows an employee to connect their laptop/PC, smart phone and tablet.
- Visitors will be granted 2 connections to allow for smart phone and laptop
- Servers will be granted 1 LAN connection
- CCTV devices will be granted 1 LAN connection per device
- IoT devices will be granted 1 LAN connection per device
- Printers will be granted 1 LAN connection per printer
- Video conferencing units will be allocated 2 LAN connections per device as they can normally have a controller on the desk which is powered via POE or higher as well as a LAN connection the video conferencing bar.

Considering how business is going and the number of regular visitors at each office, it is recommended to have at least 4 spare ports per office as this can allow for troubleshooting as well as adding additional devices while waiting for a new switch to be delivered and configured. If a new switch is configured, the same model and configuration is recommended as this will enable easy setup and configuration as well as consistent security and reliability throughout the business.

As the switches are stackable, the number of switches required will depend on number of office workers, required number of end devices at each office plus the expected number of manual workers and visitors that may visit an office however for this network demonstration, we will use 1 device per office plus 2 switches in the main 2 buildings (Brisbane and SITEA Office).

Access Points (APs)

To match the other hardware, the Cisco Wireless 9176I Access Points (APs) have been chosen as they are reliable APs with Wi-Fi 7 (802.11be) speeds as well as being backward compatible and able to connect to devices that run any of following Wi-Fi standards 802.11a/b/g/n/ac/ax/be.

The data sheet for these APs can be found in the following link followed by a picture of the device in figure 4. <https://www.cisco.com/c/en/us/products/collateral/wireless/wireless-9176-series-access-points-ds.html#Productoverview>

For reference on how to install the AP's, the cisco deployment guide for the APs is found here https://www.cisco.com/c/en/us/td/docs/wireless/access_point/technical-reference/cw9176-dg.html



Figure 4 – Cisco Wireless 9176I Access Point

Other features that may be of use to enhance features of other end devices like IoT devices include the Integrated Ultra-Wideband (UWB) radio, Intelligent Capture and Application Hosting however these features have no impact on network design and security and are therefore out of scope of this project.

The APs can be powered by the designated switch via UPOE and will have a link speed of 1x10Gbps (dependent on cable used) to enable Wi-Fi users a fast, reliable network connection. They can also be configured and managed by the Cisco Meraki Dashboard online if they will be managed by remote I.T. or by the Cisco Catalyst 9800 Series Wireless LAN Controller, granting the business flexibility in their overall design.

Considering the material offices are made of, layout, size and density of the offices in Brisbane and at site, the exact number of APs allocated per office is out of scope of this project. To display the APs working in the virtual environment (which can't emulate SSIDs), VLAN's have been assigned to ports on the AP to emulate a Wi-Fi device connecting to it.

Connections to the AP have been demonstrated below by running iperf3.

End Devices

It is assumed that all end devices are no more than 10 years old and therefore will all have RJ45 or SPF ports or Wi-Fi 5 (802.11ac) capabilities and therefore will be able to connect to the network. Therefore, the end-devices mentioned like server, PC, laptop, tablet, smart phone, IoT device, printer, CCTV camera, VC unit will have no impact on the network and therefore out of scope of this project.

Additional Hardware for Network devices

The ISO27001:2022 states in Annex A, section 8.2¹ that

"Networks and network devices shall be secured, managed and controlled to protect information in systems and applications."

Locking network equipment inside network cabinets/racks and (if possible) within locked communications room will also be recommended. The size of the network cabinets/racks will depend on the office it is in as well as any other equipment that may be stored in there. Other equipment can be like servers and UPS devices and any other IT equipment that may need to be powered by the UPS.

Locking network equipment is not only an ISO27001:2022 compliance, but it is also another defence in cyber security by physically limiting access to the network and minimizing the threat from what is commonly called the 'insider threat'.

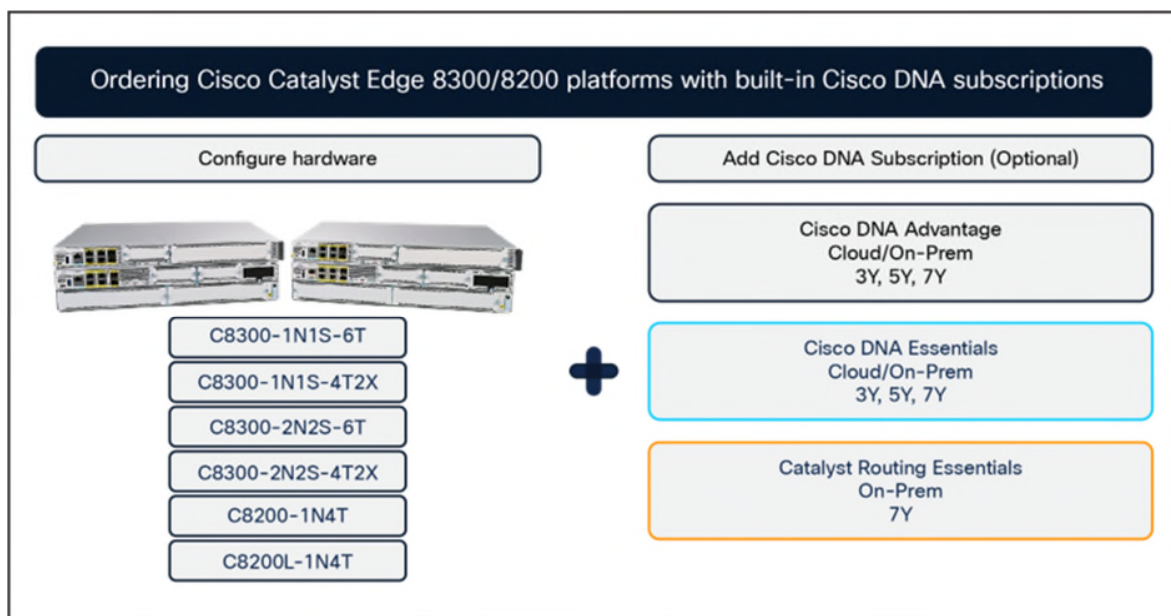
The physical security of the network devices is outside of this project scope however it will be assumed that will be part of the recommendation to PCM management.

Hardware and License Costs

Cisco devices can come with additional features if extra licenses are paid for however for a mining site, Next Business Day (NBD) is recommended as it keeps costs down plus it almost impossible to get a device replaced the same day as mines sites are remote. For this reason, it is recommended to have spare at least a spare of each device at site plus maybe an extra couple of APs so they can be installed quickly then restocked as this will allow the company to expand or change offices without waiting for IT hardware to arrive.

For an initial cost analysis, only the required hardware was requested and came in at \$286,028.17 with the full quote being in [Appendix A](#).

Other licencing options are in picture 5 with the Cisco DNS Software for SD-WAN and Routing Ordering Guide available here <https://www.cisco.com/c/en/us/products/collateral/software/dna-subscription-routing/guide-c07-740642.html>



Picture 5

Summary Tables

A summary of the devices and their features is below with more features taken from datasheets mentioned above.

DEVICE	FEATURES	Price
Cisco Catalyst 8300-1N1S-6T router	• 1RU size • 8GB DRAM Memory • 16GB M.2 SSD Storage • 1Gbps WAN ethernet port • 1 x SM slot • 1 x NIM slot • 4 x 1Gbps ethernet ports • 2 x 1/10Gbps SPF+ ports • Dual Power Supply • 2Gbps SD-WAN IPsec Throughput • 2Gbps SD-WAN IPsec Throughput with IQDF** (1400Bytes) • 1.8Gbps SD-WAN IPsec Throughput (IMIX)*** • 1.7Gbps SD-WAN IPsec Throughput with IQDF (IMIX) • 6000 SD-WAN Overlay Tunnels scale • 19.7Gbps IPv4 Forwarding Throughput (1400Bytes) • 1.9Gbps IPsec Throughput • 1.9Gbps IPsec Throughput (IMIX)** • 1.6M IPv4 Routes • 1.5M IPv6 Routes • 1.2M NAT sessions	\$20,758.28.00 + \$1,368.45 Cisco Smartnet Parts only support /device**
Cisco Secure Firewall 1220CX	• Compact/Desktop design • 8 x (10/100/1000Mbps) ethernet ports • 2 x SPF+ 1/10Gbps ports • 1.4kg weight • Dimensions = 2.82 x 27.43 x 17.22cm • Operating Temperature 0 – 40 °C	\$7,701.34.00 + \$745.35 Cisco Smartnet Parts only support /device**

	• Power consumption – 32W typical & 40W maximum • 9Gbps Firewall Throughput • 9Gbps Threat Defence • 9Gbps IPS**** • 1,5Gbps TLS***** • Performance depends on features activated, network traffic protocol mix and packet size characteristics	
Cisco Catalyst C9300-24UX	• 24 x 100M/1G/2.5G/5G/10G ethernet ports with UPOE • 8GB DRAM • Up to 4094 VLAN IDs • 480Gbps stacking bandwidth • Maximum of 8 stackable switches • 32,000 maximum MAC addresses • 32,000 Total IPv4 routes (24,000 direct and 8,000 indirect routes) • 16,000 IPv6 routing entries • 8,000 multicasting routing scale • 640Gbps switching capacity • 1120Gbps switching capacity with stacking • 476.19Mpps forwarding rate • 833.33Mpps forwarding rate with stacking	\$19,319.59 + \$2,136.39 Cisco CX Support / device**
C9300-NM-8X	• 8 x 1G/10G SFP+ ports	
PWR-C1-1900WAC-P	• Dual power supply • 1900W power supply	
Cisco Wireless 9176I Access Points (AP)	• Dimensions – 24 x 24 x 5.1cm • 1.56kg	\$3,082.91 + \$135.62 Meraki Support RMA only / device**

	• Operating temperature 0 °C – 50 °C • Operating humidity 10% - 90% • Wi-Fi 7 and backward compatible (802.11a/b/g/n/ac/ax/be) • 100M/1G/2.5G/5G/10G ethernet ports • 39W (UPOE) maximum power consumption • AP power optimizations (AP power save mode) • 4x4 up/down links with MU-MIMO spatial streams • Integrated Ultra-Wideband (UWB) radio • Band steering • Intelligent Capture • Flexible Radio Assignment (FRA)	
Cisco GLC-LH-SMD SFP	• 1000BASE-LX/LH long-wavelength; with DOM	\$1,232.86 / device**

* Link to existing product

** Price quote is in Appendix A

** IQDF=IPsec + Quality of Service (QoS) + Deep Packet Inspection (DPI) + Flexible Netflow (FNF)

*** IMIX average packet size is 352 Bytes

**** IPS = Intrusion Prevention System

***** TLS = Transport Layer Security

Reference

1. Whiting, J 2025, ISO 27001:2022 Annex A 8.20 – network security,
<https://www.isms.online/iso-27001/annex-a-2022/8-20-network-security-2022/>

Appendix A

Cisco quote

Prepared for **Duane Van Itallie**

Valid until **13-10-2026**

				\$258,525.61
Product		Quantity	Price	Amount
 Cisco Catalyst 8300 C8300-1N1S-6T Router		2	\$20,758.28	\$41,516.56
 Cisco Smartnet (Con-Snt-C83001n1) Parts Only 8X5XNBD For C8300-1N1S-6T-V		2	\$1,368.45	\$2,736.90
 Cisco Secure Firewall 1220 Appliance, Compact, Threat Defense		2	\$7,701.34	\$15,402.68
 Cisco Smartnet (Con-Snt-Csf1xtdk) Parts Only 8X5XNBD For CSF1220CX-TD-K9		2	\$745.35	\$1,490.70
 Cisco Catalyst 9300 C9300-24UX 24 Ports Manageable Ethernet Switch - Gigabit Ethernet - 10/100/1000Base-T		7	\$19,319.59	\$135,237.13
 Cisco CX Support (Con-L1nbd-C93024xu) CX L1 8X5XNBD For C9300-24Ux-A		7	\$2,136.39	\$14,954.73
 Cisco Wireless 9176I(W73 Radio3 Band 4X4		7	\$3,082.91	\$21,580.37
 Cisco Meraki Support (Con-Rob-Cw917irt) Rma Only 8X5XNBD For For Cw9176i-Rtg		7	\$135.62	\$949.34
 Cisco GLC-LH-SMD SFP (mini-GBIC) - 1 x LC/PC Duplex 1000Base-LX/LH Network		20	\$1,232.86	\$24,657.20

Subtotal	\$258,525.61
Shipping	\$1,500.00
GST	\$26,002.56
Total	\$286,028.17

All quotes are valid for 7 days unless otherwise specified. The quoted items are subject to change without notice due to rising component costs and will reflect current market rates which may be higher and stock levels that fluctuate constantly. Expired quotes cannot be refreshed at original pricing. Please Approve orders within the validity period to secure your price.

Please note installation date, run-ups, and timeframes involved with any orders and deliveries. Once a quote is approved we proceed to place the order with the distributor. If they have stock in Melbourne or Sydney it will take 2-3 business days from the date of order to reach our office. Orders placed after 10am are generally shipped the following business day in which case will take longer. Despite our best efforts to make ETAs as accurate as possible, expected dates are estimates only and are not guaranteed. They are subject to change without notice.

The shipping cost included in this quote is related to delivery from the supplier to our office, further shipments can incur an additional courier charge. Once goods have been received in our office our technician team generally takes 1-2 days to run-up and then ship your goods to you.

Payment Terms: Trading terms are 14 days from the invoice date. Orders over \$10,000 require a negotiated amount upfront as a deposit.

NOTE: Labour quotes are an estimate based on previous experiences. From time-to-time machines and systems can vary slightly which in turn will cause variations in Labour. In addition, any non-scoped work requested will be charged at the agreed rate and billed as part of this job or project. If testing determines that additional equipment is required there will be additional labour charges as part of a return trip to satisfy the requirements. If less time is spent, you will only be charged for the work that was completed.

Appendix B

Cisco Secure Firewall 1220CX Software Features

Table 6. Performance - 1200 Series with Threat Defence software

Metric	1220
	9.0
Throughput: FW + AVC (1024B)	Gbps
Throughput: Next-Generation Intrusion Prevention System (NGIPS) (1024B)	9.0 Gbps
	9.0
Throughput: FW + AVC + IPS (1024B)	Gbps
IPSec VPN Throughput (1024B TCP w/Fastpath)	10 Gbps
	1.5
TLS decryption ^[1]	Gbps
Maximum new connections per second, with AVC	50K

Table 7. Performance - 1200 Series with ASA software

Metric	1220
Stateful inspection firewall throughput ^[2]	15 Gbps
Stateful inspection firewall throughput (multiprotocol) ^[3]	12 Gbps
IPsec VPN throughput (450B UDP L2L test)	12 Gbps
New connections per second	250K

Note: Performance will vary depending on features activated, network traffic protocol mix, and packet size characteristics. Performance is subject to change with new software releases. Consult your Cisco representative for detailed sizing guidance.

Scalability

1200 Series appliances are designed for deployment in branch offices and smaller sites.

Table 8. Scalability - 1200 Series with Threat Defence software

Measure	1220
Maximum concurrent sessions, with AVC	300K
Maximum VPN Peers	300

Measure	1220
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Maximum virtual router instances (VRF) 10

Table 9. Scalability - 1200 Series with ASA software

Measure	1220
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Concurrent firewall connections 300K

Maximum VPN Peers	300
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1200 Series firewalls are powered by an advanced processor that accelerates security and networking functions.

Note: Performance will vary depending on features activated, network traffic protocol mix, and packet size characteristics. Performance is subject to change with new software releases. Consult your Cisco representative for detailed sizing guidance.